Customer Shopping Behavior Analysis

-By Suraj Jadhav

Insights from 3,900 transactions across product categories to guide strategic business decisions.

Made with GAMMA

Dataset Overview

3,900

Transactions

Purchase records analyzed

18

Features

Data points per transaction

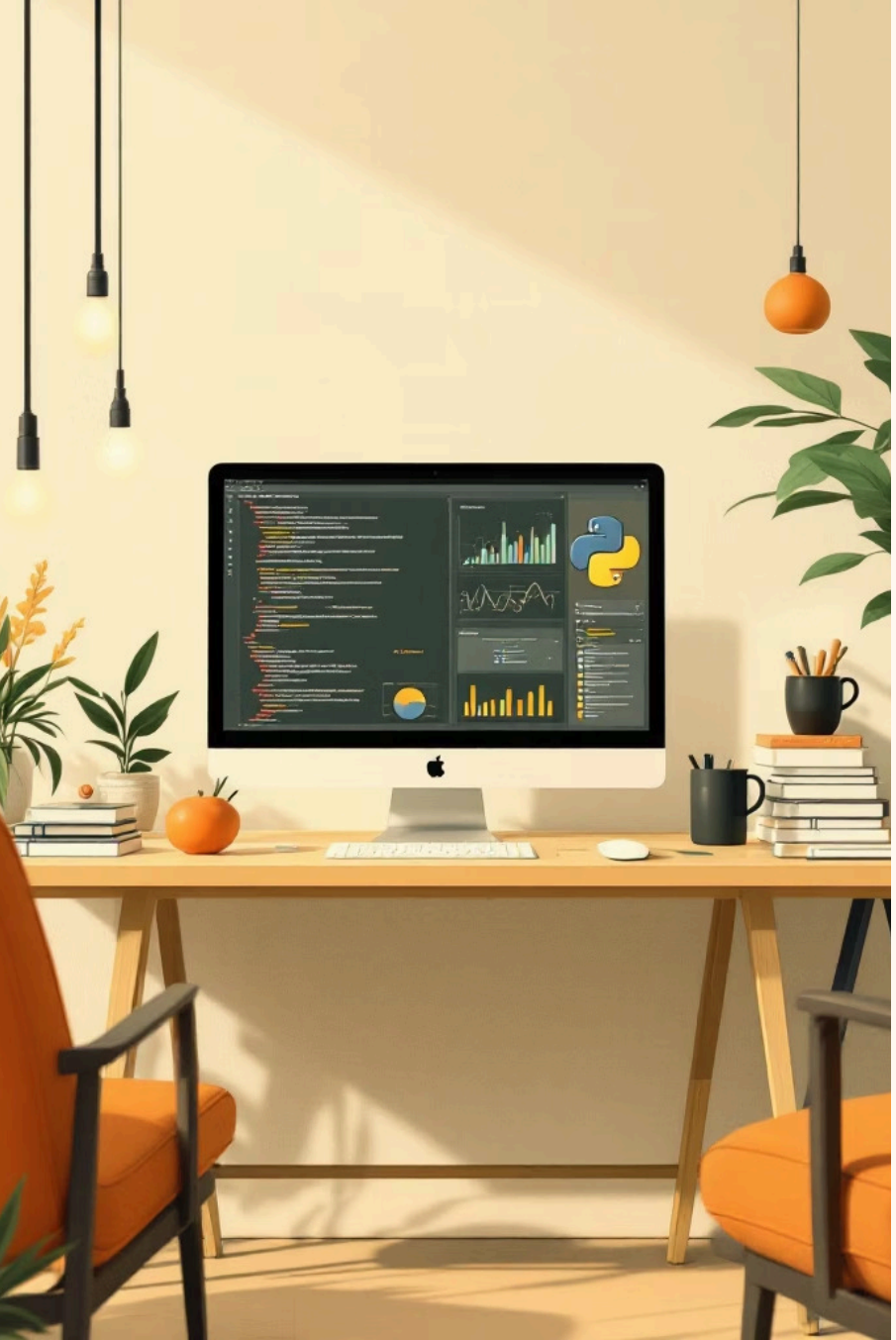
37

Missing Values

Imputed in review ratings

Key features include customer demographics, purchase details, and shopping behavior metrics.

Made with GAMMA



Python Data Preparation

Data Loading & Exploration

Imported dataset using `pandas` and conducted initial exploration with `df.info()` and `.describe()` for structure and summary statistics.

Data Cleaning

Handled missing values in `Review Rating` using median ratings by product category. Standardized column names to snake case for readability.

Made with GAMMA

Feature Engineering

01

Age Group Binning

Created `age_group` column by categorizing customer ages into segments.

02

Purchase Frequency

Generated `purchase_frequency_days` from purchase data for behavioral analysis.

03

Data Integration

Connected to PostgreSQL and loaded cleaned DataFrame for SQL analysis.

Made with GAMMA

SQL Analysis: Revenue Insights

Gender Revenue Split

Male customers generated **$157,890** vs. Female **$75,191**

Age Group Performance

Young Adults led with **$62,143** revenue, followed by Middle-aged ($59,197), Adult ($55,978), and Senior ($55,763)

Shipping Comparison

Express shipping averaged **$60.48** vs. Standard **$58.46**

Made with GAMMA

Product Performance

Top-Rated Products

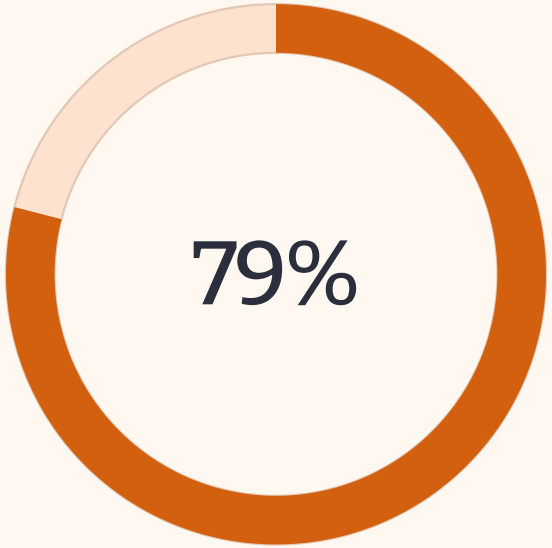
- Gloves: 3.86 rating
- Sandals: 3.84 rating
- Boots: 3.82 rating
- Hat: 3.80 rating
- Skirt: 3.78 rating

Discount-Heavy Products

- Hat: 50.00% discounted
- Sneakers: 49.66% discounted
- Coat: 49.07% discounted
- Sweater: 48.17% discounted
- Pants: 47.37% discounted

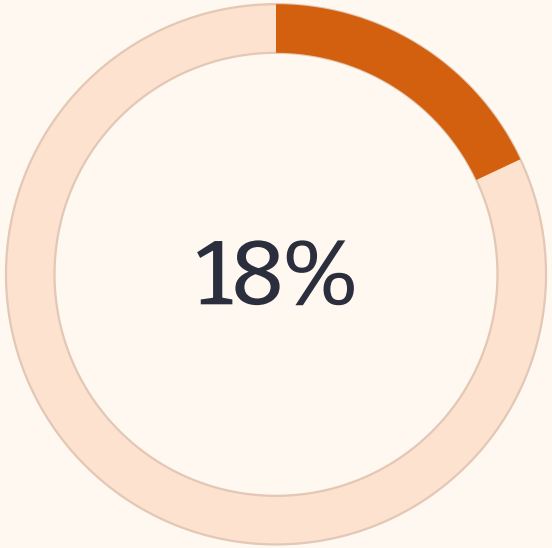
Made with GAMMA

Customer Segmentation



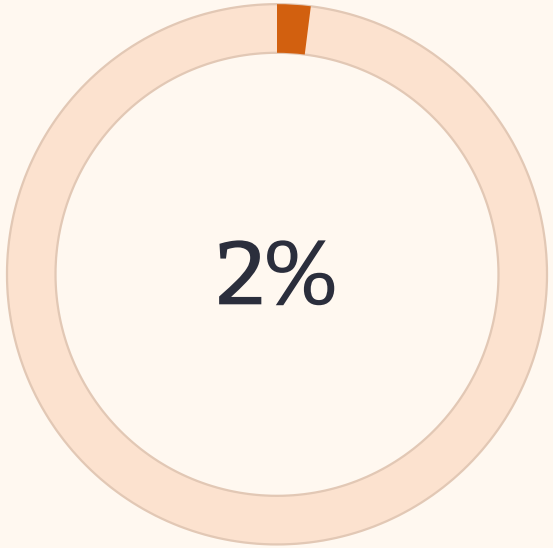
Loyal Customers

3,116 customers with consistent purchase history



Returning

701 customers making second purchases



New

83 first-time buyers in the dataset

Analysis revealed 958 repeat buyers (over 5 purchases) with subscription status tracked.

Made with GAMMA

Subscription Analysis

Subscriber Base

1,053 subscribers (27% of customers) generating $62,645 total revenue with average spend of $59.49

Non-Subscriber Base

2,847 non-subscribers (73% of customers) generating $170,436 total revenue with average spend of $59.87

Key Finding

Non-subscribers spend slightly more per transaction but generate significantly higher total revenue due to larger customer base

Made with GAMMA

Power BI Dashboard

Key KPIs

Average purchase: $59.76 | Average rating: 3.75 | 3,900 customers

Visualizations

Revenue by category and age group, subscription status donut chart, sales trends

Made with GAMMA

Strategic Recommendations

Boost Subscriptions

Promote exclusive benefits to convert non-subscribers

Loyalty Programs

Reward repeat buyers to move them into loyal segment

Discount Policy

Balance sales boosts with margin control

Targeted Marketing

Focus on high-revenue age groups and express-shipping users

Made with GAMMA